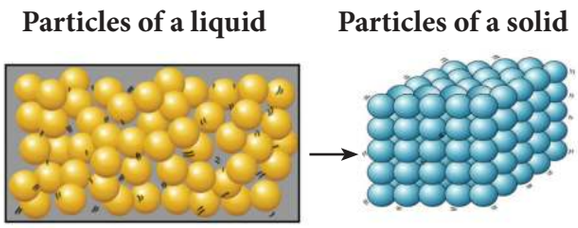
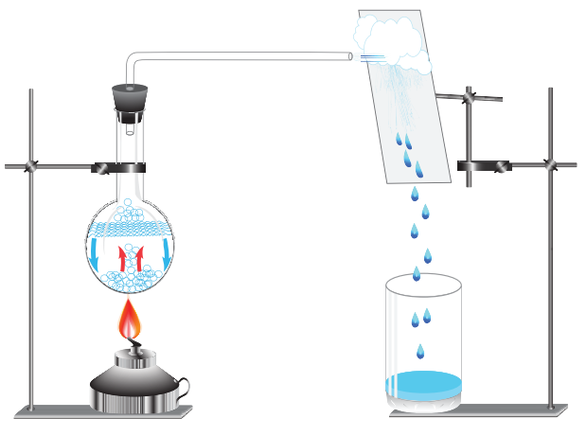
The arrangement of particles in liquid and solid are diagrammatically represented as follows:




### 3.3.4 Condensation

We would have observed that the plate that covers the cooked food items have water droplets inside. Why?



The water vapour emerges from the hot food and goes up. The plate covering the food item is in relative less temperature than the hot food. Thus the more energetic molecules loose energy once they touch the cooler plate. As the molecules lose heat, they lose energy and slow down. They move closer to other gas molecules. Finally these molecules collect together to form a liquid. Condensation happens when molecules in a gas cool down.

In class six, you learnt about water cycle in which you already know how the clouds are formed from water vapour. Water vapour condenses to form clouds.

Condensation is the conversion of gas into its liquid state. The liquid obtained after condensation can be converted back into gas on heating. So, condensation is also a physical process. During this process, only the arrangement of molecules changes from the gaseous state to liquid state. So, condensation is a physical change.

$$\text{Gas} \xrightarrow{\text{cooling}} \text{Liquid}$$

$$\text{Liquid} \xrightarrow{\text{heating}} \text{Gas}$$

**Condensation** is the changing of a gas into its liquid state and it happens by cooling, whereas **Evaporation** is the changing of a liquid into its gas state and it happens by heating.

### 3.3.5 Sublimation

We have seen camphor being burnt at home, kept in rooms to prevent entry of mosquitoes. Have you ever noticed camphor becoming liquid at any point of time? ..... It will not.

There are certain solid substances like camphor, naphthalene that get converted into gas directly upon heating without becoming liquid. **This process in which a solid is converted directly into gas is called sublimation**.

In each of the above said processes, there is a change of state due to change in temperature. But there is no change in chemical composition. By changing the temperature all these changes can be reversed. We know that change of a physical state is only a physical change. So,



VII Std Science Term-2 EM Unit 3.indd 43 27/06/2023 13:00:32